



① **Chk 03** **M.M MATRICES** *Matrices*

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Definition:
 Any rectangular arrangement of real or complex nos. subject to certain rules of operations is called a 'Matrix'. If there are 'm' rows & 'n' columns then order of matrix is $m \times n$

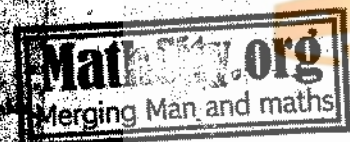
$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

→ Rows
↓ Columns Tabular Form

In abbreviated form, $A = [a_{ij}]_{m \times n}$ $i = 1, 2, 3, \dots, m$
 $j = 1, 2, 3, \dots, n$

Types Of Matrices:-

Square Matrix If $m = n$ i.e. No of Rows = No of Columns



e.g. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$

Rectangular Matrix
 If $m \neq n$

Diagonal Matrix

If all entries of a square matrix are zero except main diagonal entries.

e.g. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ $a_{ij} = 0$ when $i \neq j$
 $a_{ij} \neq 0$ when $i = j$

Identity Matrix

If all entries of a square matrix are zero except main diagonal entries which are '1'.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Zero or Null Matrix

If all entries of a matrix are zero. $O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Scalar Matrix

The matrix obtained by multiplying a non-zero scalar to each of its entries is called Scalar Matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad KA = \begin{bmatrix} Ka_{11} & Ka_{12} & Ka_{13} \\ Ka_{21} & Ka_{22} & Ka_{23} \end{bmatrix}$$

Upper Triangular Matrix: A square matrix whose elements below the main diagonal are all zero.

$$A = \begin{bmatrix} 1 & 7 & 9 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

Upper Triangular Matrix

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 7 & 5 \end{bmatrix}$$

Lower Triangular Matrix

Lower Triangular Matrix: A square matrix whose elements above the main diagonal are all zero.

Matrix Addition: A & B are conformable for addition if

$$\text{Order of } A = \text{Order of } B$$

$$A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix Multiplication: A & B are conformable for multiplication if

$$\text{No of Columns of } A = \text{No of Rows of } B$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

2x3

$$B = \begin{bmatrix} 7 & 8 & 9 \\ 0 & 5 & 1 \\ 1 & 2 & 6 \end{bmatrix}$$

3x3

AB is conformable for multiplication

$$2 \times 3 \quad 3 \times 3$$

same AB

BA is not conformable for multiplication

$$3 \times 3 \rightarrow \text{Not BA}$$

2x3

$$AB = \begin{bmatrix} 1 \times 7 + 2 \times 9 + 3 \times 1 & 1 \times 8 + 2 \times 0 + 3 \times 2 & 1 \times 4 + 2 \times 5 + 3 \times 6 \\ 4 \times 7 + 5 \times 9 + 6 \times 1 & 4 \times 8 + 5 \times 0 + 6 \times 2 & 4 \times 4 + 5 \times 5 + 6 \times 6 \end{bmatrix}$$

$$= \begin{bmatrix} 7+18+3 & 8+0+6 & 4+10+18 \\ 28+45+6 & 32+0+12 & 16+25+36 \end{bmatrix}$$

$$AB = \begin{bmatrix} 28 & 14 & 32 \\ 79 & 44 & 77 \end{bmatrix}$$

2x3

Theorem - If the matrices A, B and C are conformable for the indicated sums and products then

$$i) A(BC) = (AB)C$$

Associative Law

$$ii) A(B+C) = AB+AC$$

Distributive Law

$$iii) (A+B)C = AC+BC$$

$$iv) K(AB) = (KA)B = A(KB)$$

{ We prove the theorems by showing that element in the i th row & j th col of L.H.S. = the element in the i th row & j th col of R.H.S. }

Proof (i) $A(BC) = (AB)C$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

Let $A = [a_{ij}]_{m \times n}$ $B = [b_{ij}]_{n \times p}$ $C = [c_{ij}]_{p \times q}$

$$= \begin{bmatrix} a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31} & a_{11}b_{12}+a_{12}b_{22}+a_{13}b_{32} & a_{11}b_{13}+a_{12}b_{23}+a_{13}b_{33} \\ a_{21}b_{11}+a_{22}b_{21}+a_{23}b_{31} & a_{21}b_{12}+a_{22}b_{22}+a_{23}b_{32} & a_{21}b_{13}+a_{22}b_{23}+a_{23}b_{33} \end{bmatrix}$$

Order of $A = m \times n$

Order of $BC = \begin{pmatrix} n \times p \\ p \times q \end{pmatrix} = n \times q$

Order of $A(BC) = m \times \begin{pmatrix} n \\ n \times q \end{pmatrix} = m \times q$

Similarly

Order of $AB = \begin{pmatrix} m \times n \\ n \times p \end{pmatrix} = m \times p$

Order of $C = p \times q$

Order of $(AB)C = \begin{pmatrix} m \times p \\ p \times q \end{pmatrix} = m \times q$

Now i th row of A is $\{a_{i1}, a_{i2}, a_{i3}, \dots, a_{in}\}$

& j th column of BC is $\left\{ \begin{matrix} \sum_{\lambda=1}^p b_{\lambda 1} c_{\lambda j} \\ \sum_{\lambda=1}^p b_{\lambda 2} c_{\lambda j} \\ \vdots \\ \sum_{\lambda=1}^p b_{\lambda p} c_{\lambda j} \end{matrix} \right\}$

$\therefore$ the element in the i th row & j th col of $A(BC)$ is

$$a_{i1} \left(\sum_{\lambda=1}^p b_{\lambda 1} c_{\lambda j} \right) + a_{i2} \left(\sum_{\lambda=1}^p b_{\lambda 2} c_{\lambda j} \right) + \dots + a_{in} \left(\sum_{\lambda=1}^p b_{\lambda n} c_{\lambda j} \right)$$

$$= \sum_{\mu=1}^n a_{i\mu} \left(\sum_{\lambda=1}^p b_{\mu\lambda} c_{\lambda j} \right)$$

$$= \sum_{\mu=1}^n \sum_{\lambda=1}^p a_{i\mu} (b_{\mu\lambda} c_{\lambda j})$$

$$= \sum_{\mu=1}^n \sum_{\lambda=1}^p (a_{i\mu} b_{\mu\lambda}) c_{\lambda j} \quad \because \text{Associative Law holds in Real Nos.}$$

$$= \sum_{\mu=1}^n (a_{i\mu} b_{\mu\lambda}) \sum_{\lambda=1}^p c_{\lambda j}$$

$$= \sum_{\lambda=1}^p \left(\sum_{\mu=1}^n a_{i\mu} b_{\mu\lambda} \right) c_{\lambda j}$$

$$= \left(\sum_{\mu=1}^n a_{i\mu} b_{\mu 1} \right) c_{1j} + \left(\sum_{\mu=1}^n a_{i\mu} b_{\mu 2} \right) c_{2j} + \dots + \left(\sum_{\mu=1}^n a_{i\mu} b_{\mu p} \right) c_{pj}$$

= Element in the i th row & j th col of $(AB)C$.

$$\text{Hence, } A(B+C) = (AB)C$$

$$A(B+C) = AB+AC$$

$$\text{Let } A = [a_{ij}]_{m \times n} \quad B = [b_{ij}]_{n \times p} \quad C = [c_{ij}]_{n \times p}$$

$$\text{order of } (B+C) = n \times p$$

$$\begin{aligned} \text{order of } A(B+C) &= m \times (n \times p) \\ &= \boxed{m \times p} \end{aligned}$$

$$\text{order of } AB = m \times (n \times p) = m \times p$$

$$\text{order of } AC = m \times (n \times p) = m \times p$$

$$\text{order of } AB + AC = \boxed{m \times p}$$

$$\text{order of } A(B+C) = \text{order of } AB + AC$$

Now

i th row of A is $\{a_{i1}, a_{i2}, \dots, a_{in}\}$

$$j^{\text{th}} \text{ column of } (B+C) \text{ is } \left\{ \begin{array}{l} b_{1j} + c_{1j} \\ b_{2j} + c_{2j} \\ \vdots \\ b_{nj} + c_{nj} \end{array} \right\}$$

The element in the i th row & j th column of $A(B+C)$ is

$$a_{i1}(b_{1j} + c_{1j}) + a_{i2}(b_{2j} + c_{2j}) + \dots + a_{in}(b_{nj} + c_{nj})$$

$$= \sum_{\lambda=1}^n a_{i\lambda}(b_{\lambda j} + c_{\lambda j})$$

$$= \sum_{\lambda=1}^n [a_{i\lambda}b_{\lambda j} + a_{i\lambda}c_{\lambda j}] \quad \because a, b, c \text{ are real nos.}$$

$$= \sum_{\lambda=1}^n a_{i\lambda}b_{\lambda j} + \sum_{\lambda=1}^n a_{i\lambda}c_{\lambda j}$$

$$= \text{element in the } i\text{th row } j\text{th col of } AB + \text{element in } i\text{th row } j\text{th col of } AC$$

$$= \text{element in the } i\text{th row } j\text{th column of } (AB + AC)$$

$$\text{Thus } A(B+C) = AB + AC$$

(iii)

$$(A+B)C = AC + BC$$

$$\text{Let } A = [a_{ij}]_{m \times n}$$

$$B = [b_{ij}]_{m \times n}$$

$$C = [c_{ij}]_{n \times p}$$

$$\text{order of } A+B = m \times n$$

$$\text{order of } (A+B)C = m \times n \times p$$

$$= m \times p$$

$$\text{order of } AC = m \times n \times p = m \times p$$

$$\text{order of } BC = m \times n \times p = m \times p$$

$$\text{order of } AC + BC = m \times p$$

$$\text{Now } i\text{th row of } (A+B) \text{ is } (a_{i1} + b_{i1}, a_{i2} + b_{i2}, a_{i3} + b_{i3}, \dots, a_{in} + b_{in})$$

$$j\text{th column of } C \text{ is } \begin{Bmatrix} c_{1j} \\ c_{2j} \\ \vdots \\ c_{nj} \end{Bmatrix}$$

The element in the i th row & j th column of $(A+B)C$ is

$$(a_{i1} + b_{i1})c_{1j} + (a_{i2} + b_{i2})c_{2j} + \dots + (a_{in} + b_{in})c_{nj}$$

$$= \sum_{\lambda=1}^n (a_{i\lambda} + b_{i\lambda})c_{\lambda j}$$

$$= \sum_{\lambda=1}^n \left(a_{i\lambda} c_{\lambda j} + b_{i\lambda} c_{\lambda j} \right)$$

$$= \sum_{\lambda=1}^n a_{i\lambda} c_{\lambda j} + \sum_{\lambda=1}^n b_{i\lambda} c_{\lambda j}$$

= element in the i th row & j th column of AC + element in the i th row & j th column of BC

= element in the i th row & j th column of $AC+BC$.

$$\text{Hence } (A+B)C = AC+BC$$

x-----x

(iv) $K(AB) = (KA)B = A(KB)$

Let $A = [a_{ij}]_{m \times n}$ $B = [b_{ij}]_{n \times p}$

order of $AB = m \times n \times p = m \times p$ order of $K(AB) = m \times p$

order of $(KA) = m \times n$ order of $KB = n \times p$

order of $A(KB) = m \times n \times p = m \times p$

order of $K(AB) = m \times p = \text{order of } K(AB) = \text{order of } A(KB)$

Element in the i th row of A is $(a_{i1}, a_{i2}, \dots, a_{in})$

Element in the j th column of B is $\begin{Bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{Bmatrix}$

Element in the i th row & j th column of $AB = \sum_{\lambda=1}^n a_{i\lambda} b_{\lambda j}$

Element in the i th row & j th column of $K(AB) = K\left(\sum_{\lambda=1}^n a_{i\lambda} b_{\lambda j}\right)$

$$= \sum_{\lambda=1}^n (K a_{i\lambda}) b_{\lambda j}$$

= element in the i th row & j th column of $(KA)B$

$$= \sum_{\lambda=1}^n a_{i\lambda} (K b_{\lambda j})$$

which is element in the i th row & j th col of $A(KB)$

$$(i) \quad (A+B)^t = A^t + B^t$$

Proof Let $A = [a_{ij}]_{m \times n}$ $B = [b_{ij}]_{m \times n}$

then $A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix}$
 $A^t = \begin{bmatrix} a & e \\ b & f \\ c & g \\ d & h \end{bmatrix}$

order of $A+B = m \times n$

order of $A^t = n \times m$

order of $(A+B)^t = \boxed{n \times m}$

order of $B^t = n \times m$

order of $A^t + B^t = \boxed{n \times m}$

L.H.S

Now the element in the i^{th} row

& j^{th} column of $(A+B)^t$

= the element in the j^{th} row
& i^{th} column of $A+B$

= the element in the j^{th} row &
 i^{th} col of A + the element in the
 j^{th} row & i^{th} col of B

= $a_{ji} + b_{ji} \quad \text{--- (1)}$

R.H.S

Now the element in the i^{th} row

& j^{th} column of $A^t + B^t$

= element in the i^{th} row & j^{th}
col of A^t + element in the i^{th}
row & j^{th} col of B^t

= element in the j^{th} row & i^{th}
column of A + element in the j^{th}
row & i^{th} column of B

= $a_{ji} + b_{ji} \quad \text{--- (2)}$

(1) = (2) Hence proved.

(ii)

Let $A = [a_{ij}]_{m \times n}$

$(A^t)^t = A$

order of $A^t = n \times m$

order of $(A^t)^t = \boxed{m \times n}$

order of $A = \boxed{m \times n}$

Now element in the i^{th} row
 j^{th} column of $(A^t)^t$

= element in the j^{th} row &
 i^{th} column of A^t

= element in the i^{th} row & j^{th}
 column of A .

$$\text{iii) } (KA)^t = KA^t$$

$$\text{Let } A = [a_{ij}]_{m \times n}$$

Now element in the i^{th} row
 j^{th} column of $(KA)^t$

= element in the j^{th} row & i^{th} column
 of KA

= element in the i^{th} row & j^{th} column
 of KA^t

$$(AB)^t = B^t A^t$$

$$\text{Let } A = [a_{ij}]_{m \times n}$$

$$B = [b_{ij}]_{n \times p}$$

$$\text{order of } AB = m \times n \times n \times p = m \times p$$

$$\text{order of } (AB)^t = [p \times m]$$

$$\text{order of } B^t = p \times n$$

$$\text{order of } A^t = n \times m$$

$$\text{order of } B^t A^t = p \times n \times n \times m = [p \times m]$$

Now element in the i^{th} row and
 j^{th} col of $(AB)^t$

= element in the j^{th} row & i^{th} column
 of AB

$$= \sum_{x=1}^n a_{ix} b_{jx} \quad \text{--- (I)}$$

RHS

and the element in the i^{th} row &
 j^{th} col of $B^t A^t$

= Sum of the products of the
 corresponding elements of i^{th}
 row of B^t & j^{th} col of A^t

= Sum of the products of the corresponding
 elements of i^{th} col of B & j^{th} row of A

$$= \sum_{x=1}^n a_{jx} b_{ix} \quad \text{--- (II)}$$

provided
 it is not zero

(I) = (II) proved